

Siddesh Patil

Kalyan, Mumbai, MH | 8652109778 | siddeshpatil020@gmail.com | [linkedin/siddesh-patil](https://www.linkedin.com/in/siddesh-patil) | [github/triumph10](https://github.com/triumph10) | [Portfolio](#)

SUMMARY

Recent Computer Science & Engineering (Data Science) graduate with Honours in AI/ML. Built production-oriented systems including solar-powered AIoT platform, hybrid RAG pipelines, and semantic search engines. Eager to contribute while growing under experienced engineers.

PROJECTS

SoilStream | *Python, scikit-learn, TensorFlow, ResNet50, ESP32, Flask, Firebase* | [GitHub](#)

- Predicted regional rainfall with 92.86% accuracy using an SVR model, outperforming LightGBM and GRU baselines enabling smarter irrigation timing during monsoon season.
- Built a ResNet50 crop-classification model (97.2% accuracy) integrated into an automated irrigation controller, reducing waterlogging risk and manual monitoring effort.
- Designed a full IoT-to-cloud pipeline (ESP32 + BME280 sensors → Firebase → mobile app) for real-time field monitoring, remote pump control, and crop disease diagnosis via the Kindwise API.

VeriDoc | *Python, FAISS, BM25, Sentence-Transformers, Mistral, MLflow, Gradio* | [GitHub](#)

- Built a hybrid retrieval pipeline (dense FAISS embeddings + sparse BM25) over local PDFs, improving answer relevance over single-method retrieval before generating cited answers with an LLM (Mistral).
- Ran exploratory analysis and evaluation on retrieval outputs using the RAGAS framework, tracking faithfulness and relevancy metrics across experiment runs with MLflow bringing a data-driven approach to LLM output quality.

Sidekik | *Flask, Python, BERT, FAISS, Elasticsearch, Supabase, PostgreSQL* | [GitHub](#)

- Built a BERT + FAISS semantic matching engine to rank freelancer-project fit, improving match relevance over keyword-only search across a corpus of freelancer profiles and project records.
- Added semantic search via Elasticsearch and integrated the Adzuna API for real-time skill-demand analytics.
- Shipped a full-stack platform end-to-end (Flask APIs, Supabase backend, marketplace UI) – from data pipeline to deployed product.

PUBLICATIONS & INTELLECTUAL PROPERTY

Research Paper - IEEE CONIT 2026 (Published)

Paper ID: 960

A Solar-Powered AIoT-Based System for Intelligent Soil Monitoring and Automated Irrigation Control

Patent Filed - Government of India

App. No: 202621042187

Smart Solar-Powered AIoT System for Precision Agriculture (SoilStream)

EDUCATION

University of Mumbai

2022 - 2026

B.E. Computer Science and Engineering (Data Science) | Honours in AI/ML

TECHNICAL SKILLS

Languages: Python, SQL, Java, C

AI/ML: NLP, RAG, Recommendation Systems, SVR, CNNs, LSTM, EDA

Frameworks & Libraries: Flask, Pandas, NumPy, Matplotlib, Sentence-Transformers, FAISS, Jupyter Notebook, MLflow

Cloud & Tools: Git, GitHub, Firebase, Supabase, Linux, AWS (fundamentals)

CERTIFICATIONS

Generative AI

Google Cloud Virtual Internship (AICTE)

Google AI/ML

Google Cloud Virtual Internship (AICTE)

Data Science Methodology

IBM Developer Skills Network